Data Communication & Computer Networks — Topic 4: Networking Devices

Practice Test Booklet — 48 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/ DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — Physical-Layer Devices (Hub, Repeater, Modem & Simple Devices)

Q1. Which device operates only at the Physical layer (Layer 1) of the OSI model?

OSI model की केवल Physical layer (Layer 1) पर कौन-सा device काम करता है?

(A) Switch **(B)** Hub
(C) Router **(D)** Bridge

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6] · Easy

Q2. Which of the following pairs lists the network devices used at the physical layer?

निम्न में से कौन-सी जोड़ी physical layer पर उपयोग होने वाले network devices बताती है?

(A) Gateway, Bridge **(B)** Router, Repeater
(C) Hub, Switch **(D)** Repeater, Hub

[Source: Random Test- Rajasthan PYQ (OCR), p12 — Q93 (options reconstructed from screenshot)] · Medium

Q3. Which of the following are network connecting devices? 1. Repeater 2. Bridge 3. Hub 4. Bluetooth

निम्न में से कौन network connecting devices हैं? 1. Repeater 2. Bridge 3. Hub 4. Bluetooth

(A) 1, 2 and 3 **(B)** 1, 2 and 4
(C) 2, 3 and 4 **(D)** 1, 3 and 4

[Source: Unit Test -2 Solutions (OCR), p11] · Medium

Q4. Which of the following statements are TRUE? 1. Switch reduces collision domain. 2. Hub creates a separate collision domain for each port. 3. Router separates broadcast domains.

निम्न में से कौन-से कथन सही हैं? 1. Switch collision domain को कम करता है। 2. Hub हर port के लिए अलग collision domain बनाता है। 3. Router broadcast domains को अलग करता है।

(A) 1 and 2 only **(B)** 2 and 3 only
(C) 1 and 3 only **(D)** 1, 2 and 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q26] · Medium

Q5. Which device combines transmission from various I/O devices into one line?

कौन-सा device विभिन्न I/O devices के transmission को एक line में मिलाता है?

(A) Modifier **(B)** Multiplexer
(C) Concentrator **(D)** Gateway

[Source: Unit Test-4 Solution (OCR), p22] · Medium

Q6. A file server is a _____ which is used to store files received from various client machines.

File server एक _____ है जिसका उपयोग विभिन्न client machines से प्राप्त files संग्रहित करने के लिए होता है।

(A) Specialized computer system **(B)** Modem
(C) Router **(D)** Transponder

[Source: Unit Test-4 Solution (OCR), p24] · Medium

Q7. In a data-entry office, computer centres on the 1st and 4th floors are connected by LANs, but signals take time to travel. Which hardware can improve the data transmission?

एक data-entry office में 1वीं और 4वीं मंज़िल के computer centres LAN से जुड़े हैं, पर signals में समय लगता है. बेहतर transmission के लिए कौन-सा hardware लगाएँ?

(A) Repeater **(B)** Router
(C) Computer **(D)** Modem

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

Q8. A device that regenerates a weak signal bit-by-bit to its original strength is a:

कमज़ोर signal को bit-by-bit पुनः मूल शक्ति तक लाने वाला उपकरण है:

(A) Repeater **(B)** Amplifier
(C) Bridge **(D)** Router

[Source: Practice — original (textbook-based)] · Medium

Q9. Which of the following devices operates only in the physical layer?

निम्न में से कौन-सा device केवल physical layer पर काम करता है?

(A) Two-layer switch **(B)** Router
(C) Repeater **(D)** Bridge

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

Q10. A device that operates at the physical layer of the OSI model is:

OSI model की physical layer पर काम करने वाला device है:

(A) Bridge **(B)** Router
(C) Repeater **(D)** All of these

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Hard

Q11. All ports of a hub belong to how many collision domains?

Hub के सभी ports कितने collision domains में आते हैं?

(A) One / एक **(B)** Equal to number of ports / Ports की संख्या के बराबर

(C) Two / दो **(D)** Zero / शून्य

[Source: Practice — original (textbook-based)] · Hard

Section B — Switch / NIC / Bridge

Q12. Which device stores MAC addresses in a CAM table?

कौन-सा device MAC addresses को CAM table में संग्रहित करता है?

(A) Hub **(B)** Router

(C) Switch **(D)** Modem

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q13. A router is connected between two switches. If a broadcast frame is sent on Switch-A, what will Switch-B receive?

दो switches के बीच एक router जुड़ा है। यदि Switch-A पर broadcast frame भेजा जाए, तो Switch-B क्या प्राप्त करेगा?

(A) The broadcast **(B)** A unicast copy

(C) Nothing **(D)** A multicast

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q14. What does a MAC address represent?

MAC address किसका प्रतिनिधित्व करता है?

(A) The location of the device in the network **(B)** The unique identity of the NIC

(C) Internet connectivity **(D)** The routing path

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q15. Which device reduces collision domains (each port has its own collision domain) but does NOT break broadcast domains?

कौन-सा device collision domains को कम करता है (हर port का अपना collision domain) लेकिन broadcast domains को तोड़ता नहीं?

(A) Switch **(B)** Router

(C) Hub **(D)** Gateway

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-6] · Medium

Q16. A switch receives a frame whose destination MAC is NOT in its MAC table. What does the switch do?

एक switch को frame मिलता है जिसका destination MAC उसके MAC table में नहीं है। Switch क्या करेगा?

(A) Send the frame to the router only **(B)** Flood the frame out of all its ports (same broadcast domain)

(C) Drop the frame **(D)** Broadcast it on the entire internet

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q17. A switch has 4 VLANs configured. How many broadcast domains exist?

एक switch पर 4 VLANs configure हैं। कितने broadcast domains होंगे?

(A) 1 **(B)** 2

(C) 4 **(D)** Depends on the router

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q18. In a router-on-a-stick setup with VLANs, which device handles VLAN tagging and which handles inter-VLAN routing?

VLANs वाले router-on-a-stick setup में VLAN tagging कौन करता है और inter-VLAN routing कौन?

(A) Switch – VLAN tags, Router – routing **(B)** Router – VLAN tags, Switch – routing

(C) Both switch and router do VLAN tagging **(D)** Neither does VLAN tagging

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q19. A packet travels: wired PC → Switch → Access Point → Wireless laptop. Which devices operate at Layer 2 only?

एक packet यात्रा करता है: wired PC → Switch → Access Point → Wireless laptop। कौन-से devices केवल Layer 2 पर काम करते हैं?

(A) Switch only **(B)** Access Point only

(C) Both switch and access point **(D)** Neither

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q20. Which of the following component is responsible for network communication in a computer system?

कंप्यूटर सिस्टम में network communication के लिए कौन-सा component ज़िम्मेदार होता है?

(A) UPS **(B)** Ethernet card (NIC)
(C) SVGA card **(D)** SMPS

[Source: Unit Test-4 Solution.pdf (OCR), p18] · Medium

Q21. LANs can be connected by which of the following devices that operate on the data link layer?

LANs को data link layer पर काम करने वाले निम्न में से किस device से जोड़ा जा सकता है?

(A) Hub **(B)** Bridges
(C) HDLC **(D)** Tunnel

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

Q22. A switch learns the port for a MAC address by:

Switch किसी MAC address का port कैसे सीखता है?

(A) Reading the source MAC of incoming frames / आने वाले frames के source MAC से **(B)** Asking the router / Router से पूछकर
(C) Reading the IP address / IP address पढ़कर **(D)** Broadcasting every frame / हर frame broadcast करके

[Source: Practice — original (textbook-based)] · Medium

Q23. VLANs on a switch are used to create:

Switch पर VLAN बनाए जाते हैं:

(A) More collision domains / अधिक collision domains **(B)** Separate broadcast domains / अलग-अलग broadcast domains
(C) More repeaters / अधिक repeaters **(D)** Faster cables / तेज़ cables

[Source: Practice — original (textbook-based)] · Medium

Q24. Compared to a hub, a switch:

Hub की तुलना में switch:

(A) Sends every frame to all ports / हर frame सब ports पर भेजता है **(B)** Sends frames only to the destination port / Frame सिर्फ़ destination port पर भेजता है
(C) Works only at layer 3 / सिर्फ़ layer 3 पर काम करता है **(D)** Does not need MAC addresses / MAC addresses की ज़रूरत नहीं

[Source: Practice — original (textbook-based)] · Hard

Section C — Router / Brouter

Q25. A router performs which of the following? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

Router निम्न में से कौन-सा कार्य करता है? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

(A) 1, 2 and 4 **(B)** 2, 3 and 4
(C) 1 and 3 only **(D)** 1, 2, 3 and 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) — Q28] · Easy

Q26. A router drops some packets deliberately due to congestion. What is this technique called?

Router congestion के कारण कुछ packets जानबूझकर छोड़ देता है। इस तकनीक को क्या कहते हैं?

(A) Delay policing **(B)** Redundancy
(C) Random Early Detection (RED) **(D)** Reverse flooding

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q27. Which device creates a separate broadcast domain?

कौन-सा device अलग broadcast domain बनाता है?

(A) Switch **(B)** Bridge
(C) Router **(D)** Repeater

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q28. Which device operates at Layer 3 of the OSI model?

OSI model की Layer 3 पर कौन-सा device काम करता है?

(A) Hub **(B)** Switch
(C) Router **(D)** Repeater

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard)] · Medium

Q29. Consider: S1: The TTL field decreases at each router. S2: The MAC address changes at each hop. Which is correct?

देखिए: S1: TTL field हर router पर घटती है। S2: MAC address हर hop पर बदलता है। कौन-सा सही है?

(A) Only S1 **(B)** Only S2
(C) Both S1 and S2 **(D)** Neither S1 nor S2

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q30. A brouter is a device that can act as:

Brouter वह उपकरण है जो बन सकता है:

(A) Hub + repeater **(B)** Bridge + router
(C) Switch + modem **(D)** Firewall + gateway

[Source: Practice — original (textbook-based)] · Medium

Q31. Which hardware forwards data packets between computer networks?

कौन-सा hardware computer networks के बीच data packets को आगे भेजता है?

(A) Switch **(B)** Router
(C) Cable **(D)** Nibble

[Source: KVS Librarian 2023 Official Paper (via Testbook) · also CN Topic Test-03] · Hard

Q32. Among these network devices — i. Repeater ii. Hub iii. Switch iv. Router — which are considered intelligent devices?

इन network devices में — i. Repeater ii. Hub iii. Switch iv. Router — कौन-से intelligent devices माने जाते हैं?

(A) i and iv only **(B)** i and ii only
(C) iii and iv only **(D)** ii and iii only

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

Q33. Consider the statements about an IP router: I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the incoming fragment. Which is correct?

IP router के बारे में कथन देखिए: I. Forwarding के दौरान router IP packets को नहीं बदलता। II. Router के लिए कोई routing protocol लागू करना आवश्यक नहीं है। III. यदि outgoing link का MTU आने वाले fragment से बड़ा हो तो router को IP fragments फिर से जोड़ने चाहिए। कौन-सा सही है?

(A) I and II only **(B)** I only
(C) II and III only **(D)** II only

[Source: Unit Test-4 Solution (OCR), p15] · Hard

Q34. When a router forwards a packet, which header field does NOT change?

Router के packet forward करने पर कौन-सा header field नहीं बदलता?

(A) TTL **(B)** Header checksum
(C) Source IP address **(D)** Both TTL and checksum / TTL और checksum दोनों

[Source: Practice — original (textbook-based)] · Hard

Section D — Gateway, Firewall & NAT Devices

Q35. Which type of firewall inspects packets at the network layer and decides based on IP addresses?

कौन-सी firewall network layer पर packets की जाँच करके IP addresses के आधार पर निर्णय लेती है?

(A) Packet Filtering Firewall **(B)** Stateful Firewall
(C) Proxy Firewall **(D)** Application Firewall

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q36. Which firewall acts as an intermediary between the client and server at the application layer?

कौन-सी firewall application layer पर client और server के बीच मध्यस्थ (intermediary) के रूप में काम करती है?

(A) Packet Filtering Firewall **(B)** Stateful Firewall
(C) Proxy Firewall **(D)** Circuit-level Gateway

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q37. If two systems use DIFFERENT protocols, which device is used to link them?

यदि दो systems भिन्न protocols उपयोग करते हैं, तो उन्हें जोड़ने के लिए कौन-सा device उपयोग होता है?

(A) Repeater **(B)** Gateway
(C) Bridge **(D)** Hub

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

Q38. Which device can operate across all 7 layers of the OSI model?

कौन-सा device OSI model की सभी 7 layers पर काम कर सकता है?

(A) Router **(B)** Hub
(C) Gateway **(D)** Switch

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q39. Devices on one network can communicate with devices on another network via:

एक network के devices दूसरे network के devices से किसके माध्यम से संवाद कर सकते हैं?

(A) File server **(B)** Utility server
(C) Printer server **(D)** Gateway

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

Q40. What is the role of a firewall in computer networks?

Computer networks में firewall की भूमिका क्या है?

(A) Managing IP addresses **(B)** Filtering and controlling network traffic
(C) Resolving domain names **(D)** Assigning MAC addresses dynamically

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

Q41. What is the purpose of a proxy server in networking?

Networking में proxy server का उद्देश्य क्या है?

(A) Managing IP addresses **(B)** Filtering and forwarding network traffic
(C) Assigning MAC addresses dynamically **(D)** Encrypting network communication

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

Q42. Which is a network security system designed to protect a trusted private network from an untrusted outside network to which it is connected?

कौन-सी network security system trusted private network को उस जुड़े हुए untrusted outside network से बचाने के लिए बनी है?

(A) Bridge **(B)** Cookies
(C) Firewall **(D)** SSL

[Source: Random Test- Rajasthan PYQ (OCR), p6 — Q110 (options reconstructed from screenshot)] · Medium

Q43. What kind of NAT maps one private IP to exactly one public IP?

किस प्रकार का NAT एक private IP को ठीक एक public IP से जोड़ता है?

(A) Dynamic NAT **(B)** Static NAT
(C) PAT (Port Address Translation) **(D)** Port Forwarding

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q44. A firewall that keeps track of the state of live connections is called:

वह firewall जो चालू connections की state याद रखता है, कहलाता है:

(A) Packet-filter firewall **(B)** Stateful firewall
(C) Repeater **(D)** Modem

[Source: Practice — original (textbook-based)] · Medium

Q45. The area where public-facing servers (web, mail) are placed for security is called:

जहाँ public servers (web, mail) सुरक्षा हेतु रखे जाते हैं, वह क्षेत्र है:

(A) LAN **(B)** VLAN
(C) DMZ **(D)** SAN

[Source: Practice — original (textbook-based)] · Medium

Q46. Which NAT type uses a single public IP for many internal hosts?

कौन-सी NAT कई internal hosts के लिए एक ही public IP प्रयोग करती है?

(A) Static NAT **(B)** Dynamic NAT
(C) Overload / PAT **(D)** No NAT

[Source: Practice — original (textbook-based)] · Medium

Q47. Which of the following is NOT a networking device?

निम्न में से कौन-सा networking device नहीं है?

(A) Gateway **(B)** Linux
(C) Router **(D)** Firewall

[Source: InstrumentationTools 150 CN MCQs] · Hard

Q48. Which of the following is NOT a networking device? (Hub / Switch / Bridge / None of these)

निम्न में से कौन-सा networking device नहीं है? (Hub / Switch / Bridge / इनमें से कोई नहीं)

(A) Hub **(B)** Switch
(C) Bridge **(D)** None of these

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Hard

Answer Key — sirf letters (no solutions)

Q1: B **Q2:** D **Q3:** A **Q4:** C **Q5:** B **Q6:** A **Q7:** A **Q8:** A **Q9:** C **Q10:** C
Q11: A **Q12:** C **Q13:** C **Q14:** B **Q15:** A **Q16:** B **Q17:** C **Q18:** A **Q19:** C **Q20:** B
Q21: B **Q22:** A **Q23:** B **Q24:** B **Q25:** A **Q26:** C **Q27:** C **Q28:** C **Q29:** C **Q30:** B
Q31: B **Q32:** C **Q33:** D **Q34:** C **Q35:** A **Q36:** C **Q37:** B **Q38:** C **Q39:** D **Q40:** B
Q41: B **Q42:** C **Q43:** B **Q44:** B **Q45:** C **Q46:** C **Q47:** B **Q48:** D